Air Conditioning – General Data

1. General

The heating and ventilating system utilizes bleed air and fresh air from a scoop intake in the fuselage to provide heated, ram, or mixed air to maintain appropriate temperatures in the cabin and flight compartment. Heated air is circulated through ducts beneath the flooring to outlets at the base of each cabin and flight compartment floor. Two tubes, one for each side of the windshield, deliver heated air to the windshield defrost outlets. The same ducts also circulate cooling ram air or a mixture of heated air and ram air in the cabin and flight compartment. Two additional tubes, one for each outboard side of the flight compartment, deliver unconditioned ram air to the gaspers located on each outboard side of the instrument panel.

Additional ducts deliver ram air to the cabin through individual passenger gaspers located in the cove moulding ducts on each side of the cabin upper walls. A fan is installed in the air intake duct for forced ram air ventilation when on ground. the system provides for manual temperature control by means of a manually controlled motorized hot air valve and manual control of both ram air inflow and cabin/flight compartment air distribution balance. An exhaust vent is installed in the cabin roof.

Refer to Figure 1 for details.

A. Heating System

Compressed and heated bleed air from the engine compressors passes through pipes to the main section of the heating system where, depending on temperature selection, it is controlled and mixed with secondary air (ram, fan pressure or recirculated air). The warm mixed air then passes through a silencer, and is ducted to outlets at floor level in the cabin, pilot's foot warmers, windshield and flight compartment outlets. (Refer to 21-40-00, Heating System – General Data). The heating system is controlled manually, the control switches being mounted on the co-pilot's sub-panel on the instrument panel in the flight compartment. (Refer to 21-60-00, Environmental Control System (Mod 6/2014) – General Data).

B. Cooling System

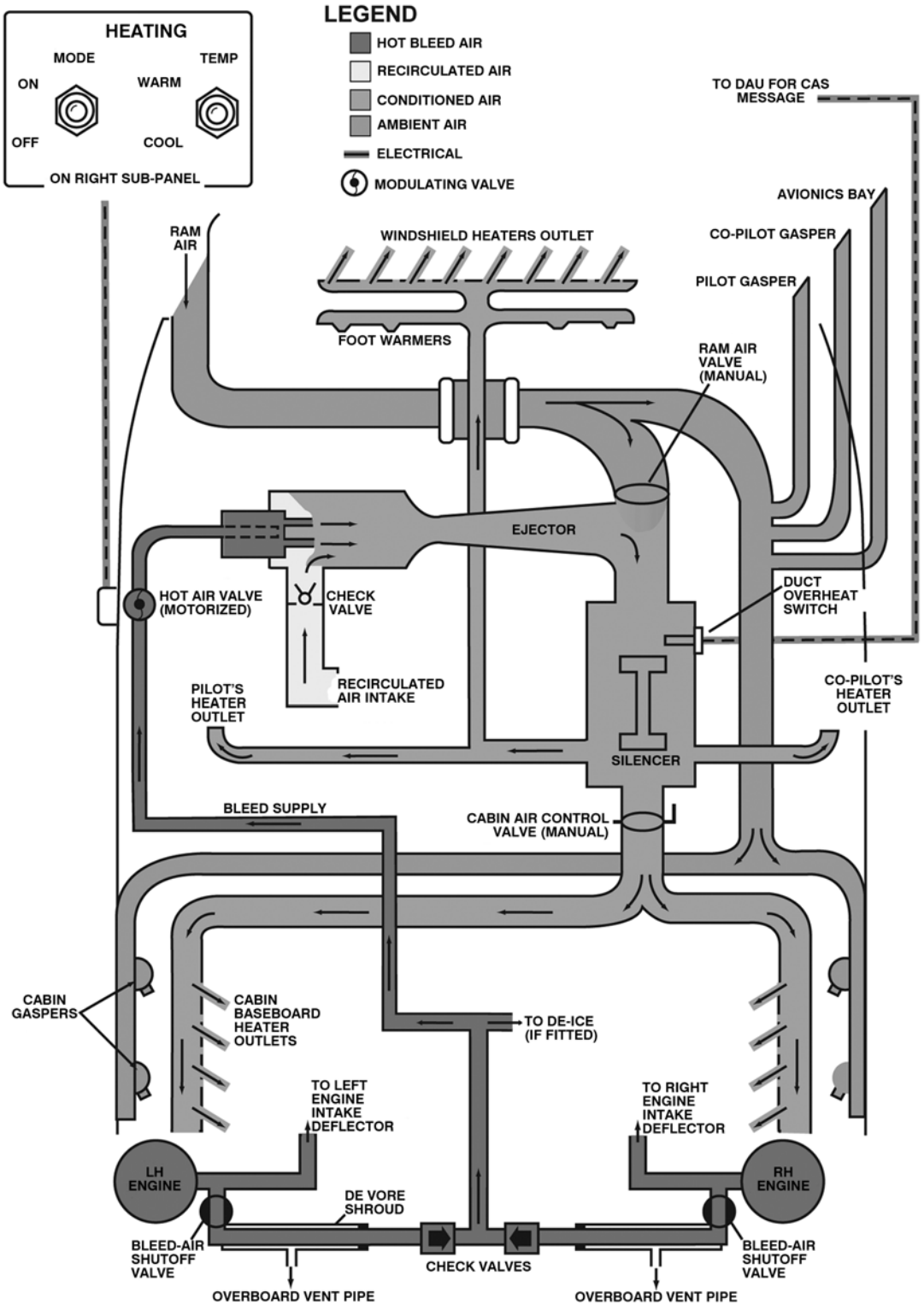
Ram air, ducted from a scoop on the left side of the fuselage nose, is supplied to the heating system, and two gaspers located on both sides of the cabin above the windows. The ram air valve, manually controlled by a handle in the flight compartment, controls the amount of ram air to the heating system. To provide a supply of cool air when the aircraft is stationary, an electric fan is installed in the main ram air duct. Additional flight compartment ventilation can be obtained by opening the flight compartment windows. Flight compartment ventilation is provided by two gaspers in the instrument panel.

C. Air Exhausting

A portion of the exhaust air vents through a vent assembly mounted in the cabin roof. The vent outlet is covered externally by a fairing which is open at the rear end. A shielded vent prevents rain entry when aircraft is parked in adverse weather conditions (Mod 6/1633). A suction, induced by the vent assembly, extracts the stale air from the aircraft interior through a grille in the cabin ceiling. Stale air also passes through vents in the luggage compartment, and leaves the aircraft at the tail cone.

D. Temperature Control System

The temperature control system controls the opening and closing of the hot air valve, installed in the heating system bleed air supply line, to maintain a selected cabin temperature. The control switches are mounted on the co-pilot's subpanel on the instrument panel in the Environmental section of the flight compartment. (Refer to 21-60-00, Environmental Control System (Mod 6/2014) – General Data).



Air Conditioning – Schematic
Figure 1